

Zomato Business Intelligence Project

Exploratory Data Analysis Report | Orders Jan 2023 - Dec 2024

1. Executive Summary

This report explores 20,475 orders, 9,254 customers and 1,200 restaurants across 25 Indian cities. All figures come from the three analytical tables built in notebook 02, and every chart is reproduced in notebook 03. The headline finding is that the business is **evenly spread but operationally leaky**: revenue has no dominant city, month or cuisine, yet a third of all orders never turn into revenue.

Headline metric	Value
Revenue (Delivered + Delivered Late orders)	Rs 5.31M
Total order value placed (all statuses)	Rs 8.02M
Average order value (delivered)	Rs 392
Cancelled / Food not delivered / Late	22.3% / 11.5% / 10.5% of orders
Late deliveries (share of delivered orders)	15.9%
Average delivery time: on-time vs late	34 min vs 64 min
Customers with only 1 order	43.9%

- Revenue is flat and evenly spread across cities, months and cuisines (Rs 5.31M over two years).
- Orders peak at lunch and dinner, but failure rates are identical in peak and off-peak hours.
- Only 55.6% of orders are delivered cleanly; 33.8% of order value is lost to cancellations and undelivered food.
- Delivery time barely responds to traffic, rain, vehicle, city or basket size - a key reason the ML models perform near chance.
- Membership tier makes no visible difference to order value or lifetime value.

Revenue definition used throughout: FinalAmount of orders with status Delivered or Delivered Late. Note that RestaurantRevenue and TotalRevenue in the analytical tables count all orders including cancelled ones.

2. Dataset Overview

Analytical table	Grain	Rows	Purpose
order_analytics.csv	1 row = 1 order	20,475	EDA, delivery-time ML, Power BI
customer_analytics.csv	1 row = 1 customer	9,254	Churn ML, customer pages
restaurant_analytics.csv	1 row = 1 restaurant	1,200	Restaurant performance page

The order table has 74 columns covering order details, customer and restaurant attributes, delivery partner, weather, traffic and 25+ engineered features (peak hour, tenure, lifetime value, delivery efficiency, rain impact and others). Customer analytics is time-aware: features use only orders on or before the observation date (2024-11-01) and the churn label looks at the following 60 days. Only customers with history before that date are included (9,254 of 12,000).

3. Data Quality and Cleaning

Cleaning followed one rule: fix what is clearly wrong, never guess what is missing. Key actions from the cleaning log:

- Removed 205 exact duplicate orders (20,705 to 20,500) and 180 duplicate customers (12,180 to 12,000).
- Dropped about 25 orders with no valid customer or restaurant link (final: 20,475 orders).

- Removed 230 contradictory duplicate feedback rows and blanked 126 impossible delivery ratings (above 5).
- Parsed mixed date formats only where unambiguous; 1,737 order dates were recovered from payment dates and 1,176 remain blank.
- Standardised city names to the 25 canonical cities; fixed negative amounts only where they could be cross-checked.

All 12 automated quality checks on the analytical tables pass (unique IDs, no negative amounts or delivery times, valid churn label, cancellation rate within 0-1). Remaining gaps are genuine source gaps, not join errors:

Column group	Missing	Reason
CouponCode	64.8%	Only orders that used a coupon have a code
Weather (rain, temperature, humidity)	about 46-47%	No reading for many city-days in the source
Traffic (level, score, speed)	about 46%	No reading for many city-days in the source
Customer / delivery / food ratings	32-33%	No feedback left for the order
OrderDate	5.7% (1,173 orders)	Ambiguous or missing in raw data; excluded from time charts

4. Revenue and Sales Insights

Delivered orders generated Rs 5.31M, almost identical in 2023 and 2024. Monthly revenue moves in a narrow band (Rs 178k to Rs 233k), with no clear seasonal pattern.

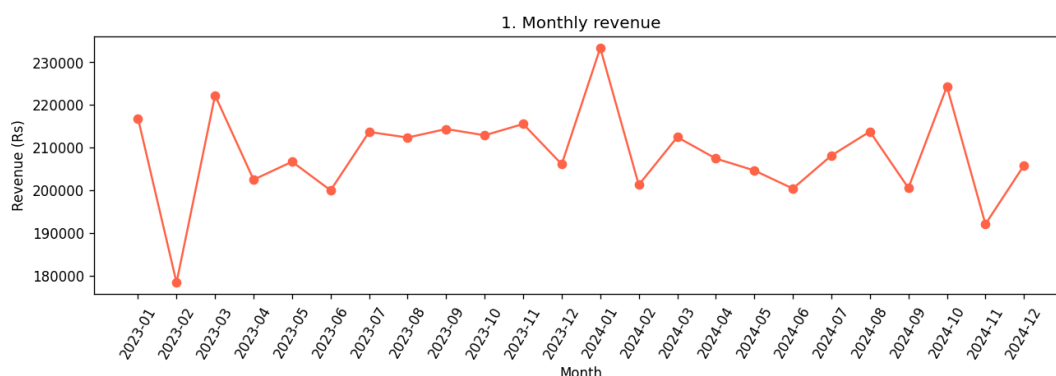


Chart 1 - Monthly revenue (orders with a valid date)

Revenue is spread across all 25 cities: the top city (Delhi) contributes Rs 247k and the 10th city about Rs 224k. Among cuisines, South Indian (Rs 0.37M) and Fast Food lead, and Thai is lowest (Rs 0.21M); differences are modest.

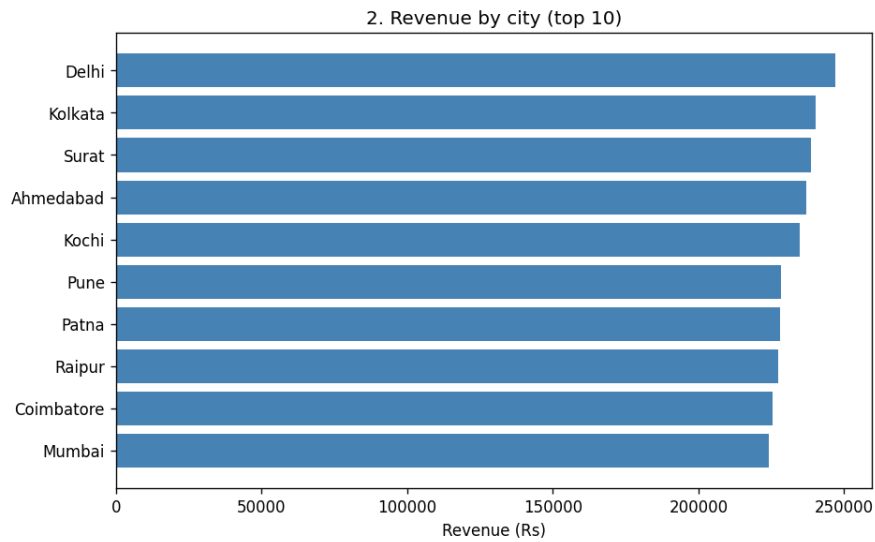


Chart 2 - Revenue by city (top 10)

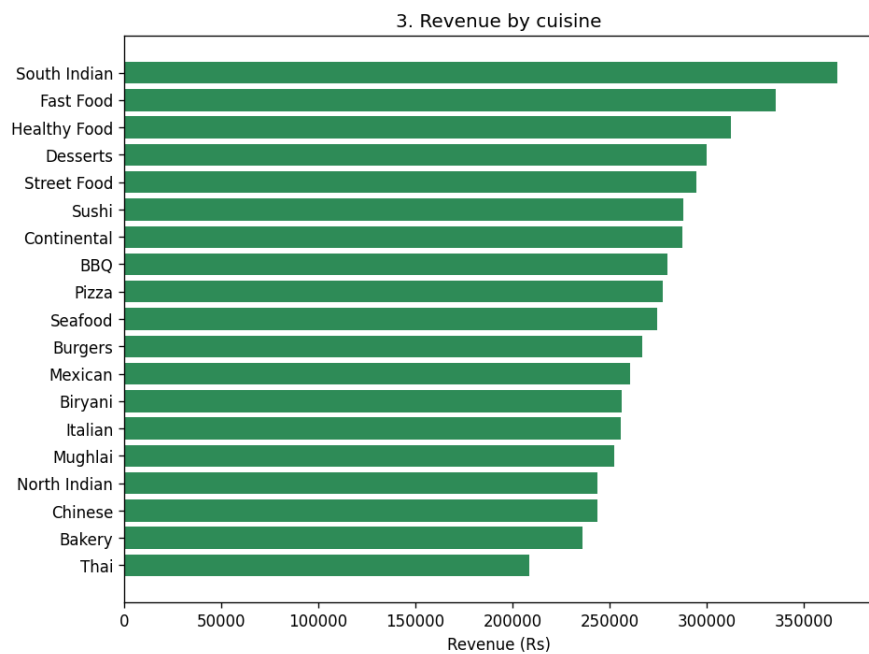


Chart 3 - Revenue by cuisine

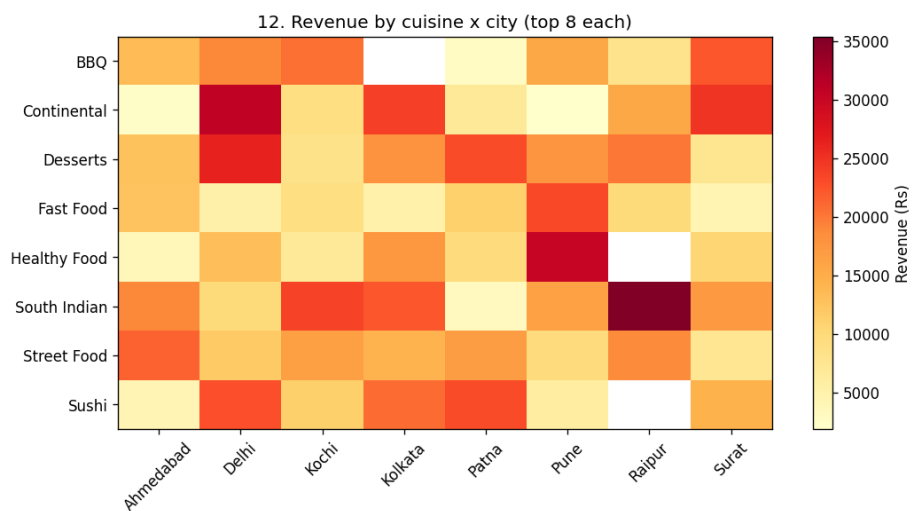


Chart 12 - Revenue by cuisine and city (top 8 each)

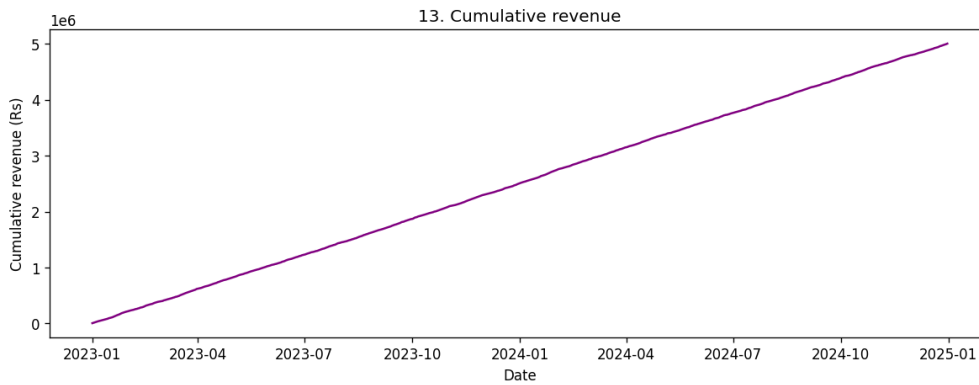


Chart 13 - Cumulative revenue

5. Order Behaviour

Orders peak at lunch (12-1 pm) and dinner (7-8 pm). The peak-hour flag (12-2 pm and 7-9 pm) covers 44.2% of all orders, and the four busiest hours alone carry 33.4%. Weekday and weekend order volumes per day are almost identical (about 26 orders per day each).

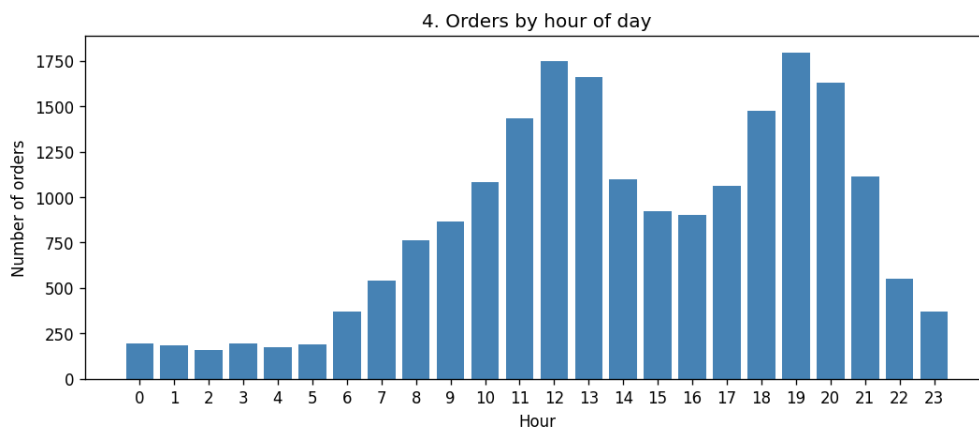


Chart 4 - Orders by hour of day

Payment methods are split almost evenly across six options (16-17% each), so no single method dominates. About 35.2% of orders use a coupon; total discounts are Rs 0.76M, or 9.7% of food cost.

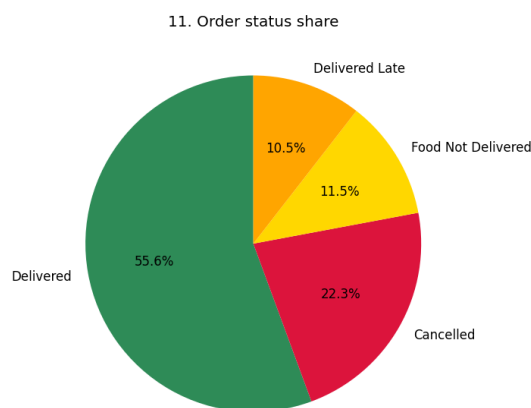


Chart 11 - Order status share

Cancellation rates run from about 19% (Chinese, Continental) to 25% (Street Food, Mexican). The gaps are modest but consistent enough to monitor.

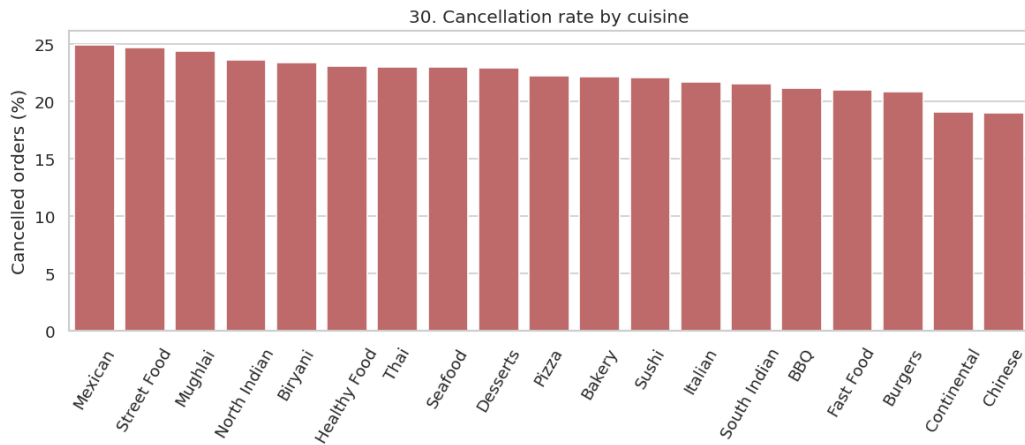


Chart 30 - Cancellation rate by cuisine

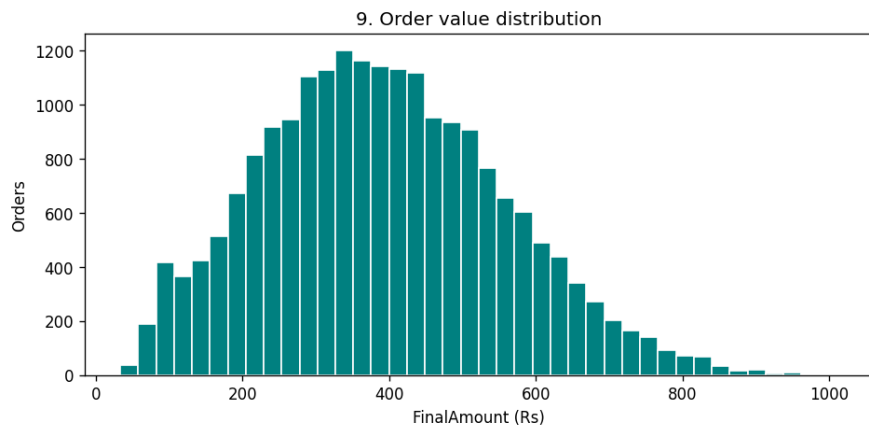


Chart 9 - Order value distribution

6. Customer Insights

Customers order rarely: the average is 1.9 orders per customer (up to the observation date), 43.9% have ordered once and 75.9% at most twice.



Chart 28 - Customers by number of orders

Membership tier makes no difference: average lifetime value is Rs 751 (Basic), Rs 747 (Gold) and Rs 760 (Zomato Pro), and average order value is about Rs 392 in every tier.

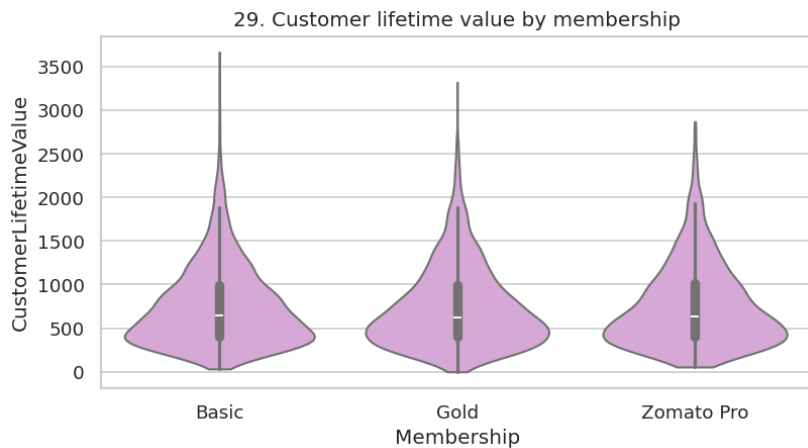


Chart 29 - Customer lifetime value by membership

7. Restaurant Insights

The 1,200 restaurants receive a similar number of orders (average 17, range 6 to 35). The top 10% of restaurants account for 14.8% of order value - barely above an even 10% split - so there is no dependency on a few star restaurants. Restaurant rating has almost no relationship with revenue (correlation -0.05) or with cancellation rate (0.02). PerformanceScore (a documented weighted blend of rating, revenue, volume and cancellation) splits restaurants into four equal quartiles.

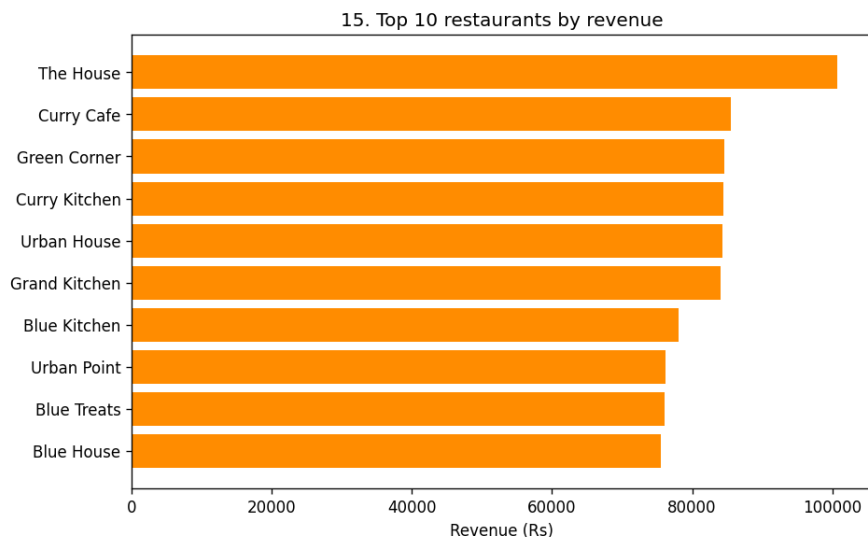


Chart 15 - Top 10 restaurants by revenue

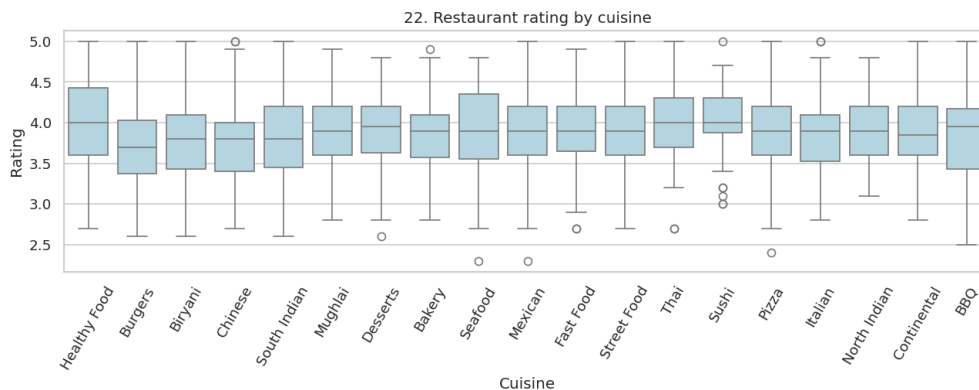


Chart 22 - Restaurant rating by cuisine

8. Delivery Insights

Delivered orders take about 38 minutes on average, but the distribution has two parts: on-time deliveries average 34 minutes and late ones 64 minutes. 15.9% of delivered orders are late.

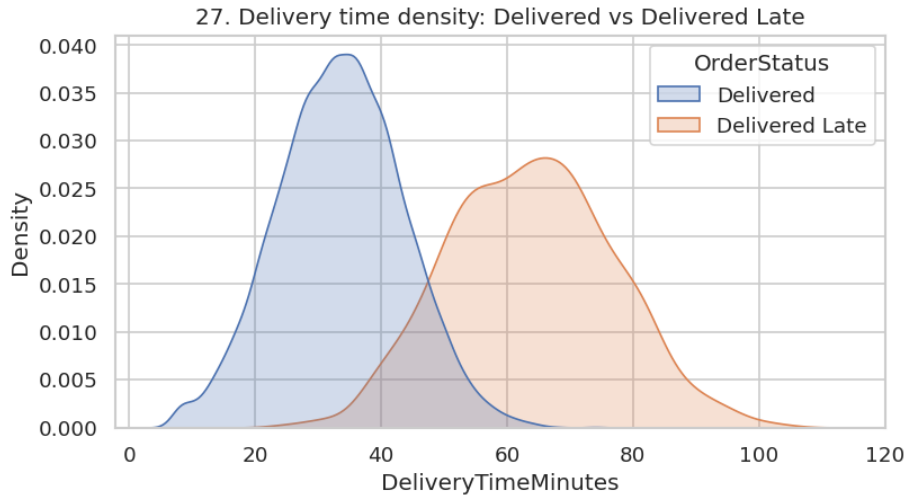


Chart 27 - Delivery time: Delivered vs Delivered Late

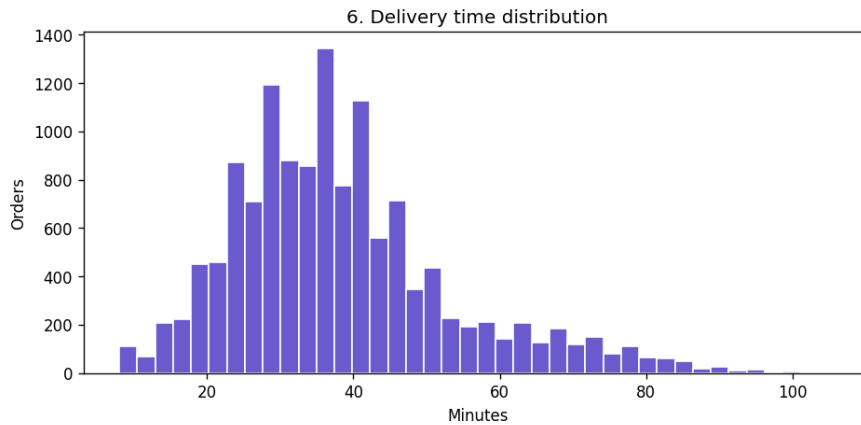


Chart 6 - Delivery time distribution

Average delivery time barely differs across cities (about 37 to 40 minutes) or vehicle types (about 38 minutes, bicycles slightly faster). Traffic has only a mild effect: late rate rises from 14.3% in low traffic to 18.0% in severe traffic.

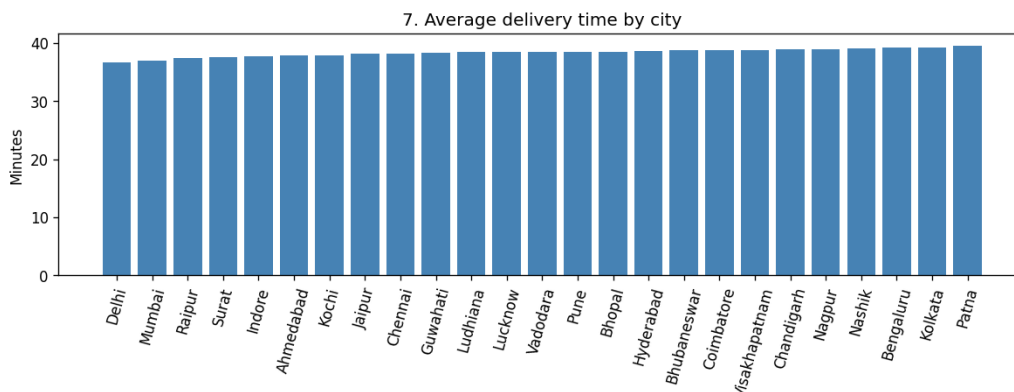


Chart 7 - Average delivery time by customer city

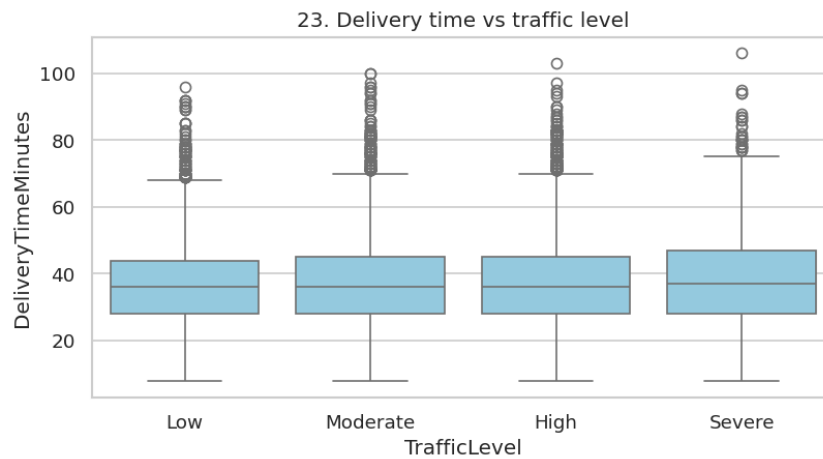


Chart 23 - Delivery time vs traffic level

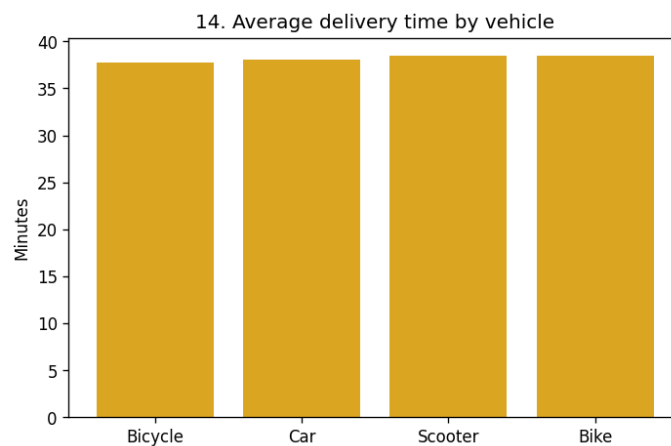


Chart 14 - Average delivery time by vehicle

9. Key Relationships

The correlation heatmap shows almost every variable is uncorrelated with delivery time (0.03 or lower). Only two relationships are strong, and both are structural: average speed falls as traffic rises (-0.68), and discount rate falls as order value rises (-0.34, because it is calculated as a share of the bill).

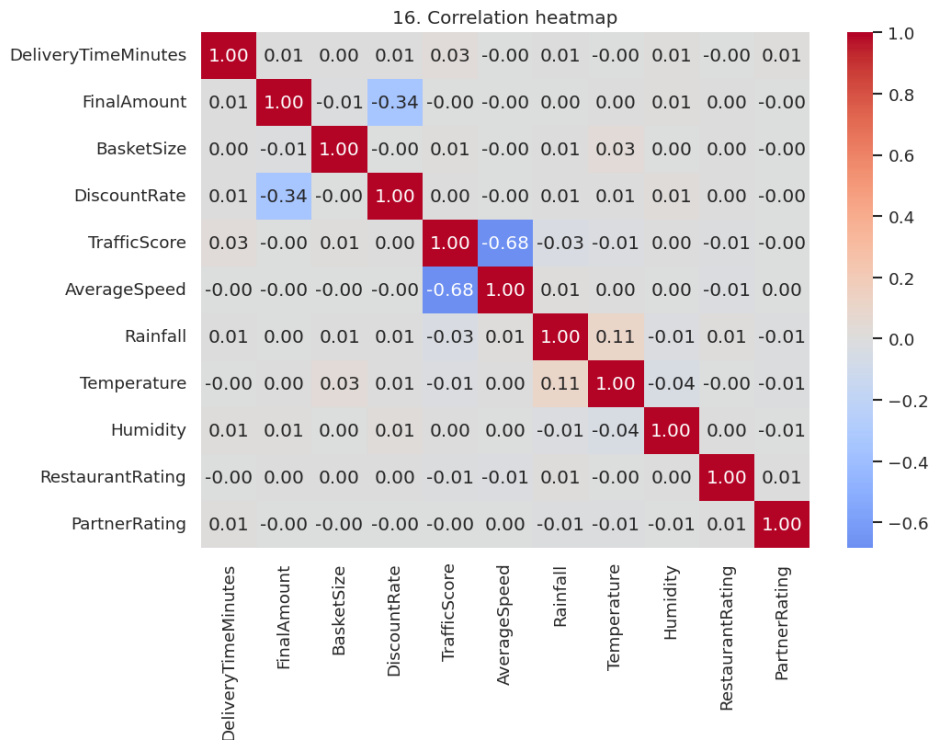


Chart 16 - Correlation heatmap

Machine-learning results are consistent with this: the delivery-time model reaches R-squared of about 0 (MAE about 11.6 minutes), and the churn model reaches ROC-AUC of about 0.50, no better than guessing. Churn is about 88% in every recency and order-count group, so the available features do not carry a usable signal.

10. Business Implications

- Reducing cancellations and undelivered orders is the largest revenue opportunity: about 34% of order value does not convert to revenue.
- Peak-hour capacity alone will not fix failures, because cancellation and lateness are the same in peak and off-peak hours.
- Late-delivery differences between cities (roughly 12% to 18%) point to city-level operations rather than a network-wide problem.
- Membership benefits are not visibly changing behaviour and should be reviewed.
- With most customers ordering once or twice, second-order conversion matters more than churn prediction.

11. Limitations

- No customer coordinates exist, so distance was not calculated (nothing was fabricated). SameCity is a weak proxy: only 4% of orders have customer and restaurant in the same city.
- About 46% of orders lack weather and traffic readings; the delivery model imputes these with medians.
- 1,173 orders without a valid date are excluded from time-based charts.
- Restaurant-level rates rest on small samples (about 17 orders each) and should be read with caution.
- Both ML models perform near chance; this reflects the dataset, not the code.